Tempo

A week you can *actually* finish.

An AI study planner that learns how long work really takes *you*, and tells you on day one when a week doesn't fit.

AI FOR STUDENT LIFE · RIT DUBAI

[Team name] · [School]

01 · THE PROBLEM

No planner tells you if your week *fits*

We underestimate

Work takes longer than we predict. Psychologists call it the planning fallacy.

Buehler, Griffin & Ross, 1994

We cram

Spreading study over days beats one long night, but the night before is what happens.

Cepeda et al., 2006

We find out too late

An impossible week looks fine on Monday. You only notice on Wednesday night.

No planner warns you

Our survey: **[X] of [Y]** students said **[“…”]**

A busy student who won't fill in a 20-field form



Grade 10–12

- 4–6 subjects
- Sports, clubs or a job
- 1–3 hours a day to study
- Plans with guesswork

They need…	So Tempo…
No forms	Reads your week as you'd text it
Real time estimates	Learns your pace per subject
Plans that keep up	Re-plans instantly, says what moved
Privacy, no Wi-Fi	Local AI and offline rules
Honesty when it's too much	Says "doesn't fit", offers options

Tell it. Check it. Let the plan keep up with you

01

Tell

Type or speak your week in plain words.

02

Check

Confirm what the AI found. Guesses are yellow.

03

Plan

Sessions around activities and sleep, each with a why.

04

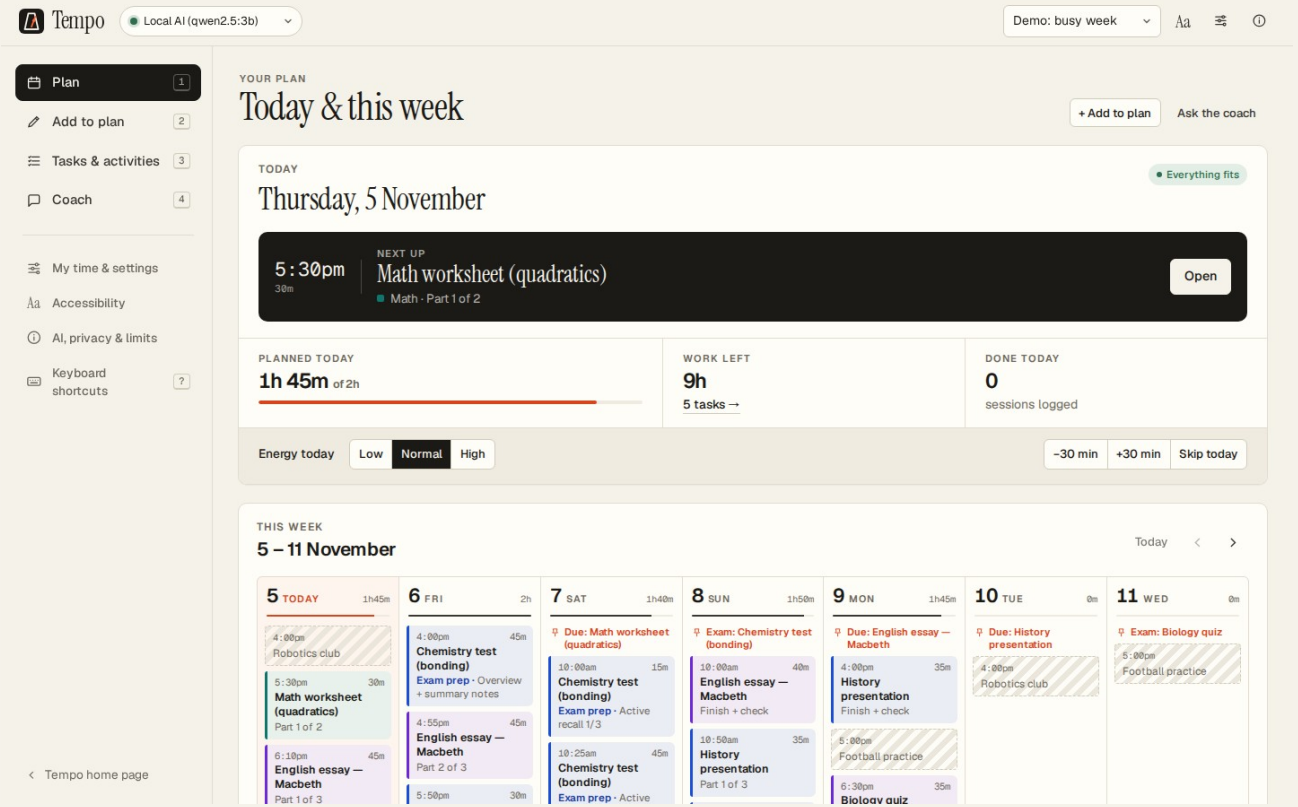
Adapt

New task, moved exam, tired today: it re-plans and says what moved.

And when the week is impossible, it says so on day one and offers options it has **already checked**.

04 · LIVE DEMO

The judges' week, planned in seconds



1. Type the week, press Understand
2. Check the yellow guesses
3. Add: see what changed
4. Open a session: why it's there
5. Log 60 min: Math estimate rises
6. Tap Low energy: today halves
7. Exam crunch: checked options

`tempo-study-planner.netlify.app/app`

05 · THE AI

Four jobs, and the right kind of AI for each

LANGUAGE MODEL

Understand what you type

Local AI or Groq turns messy text into tasks, checked by code.

STATISTICAL MODEL

Predict how long it takes you

Learns your pace per subject from the times you log.

PLANNING ALGORITHM

Plan without breaking rules

Deadlines first, spaced revision, buffers, sleep protected.

LANGUAGE MODEL + RULES

Coach you on your plan

Answers from your plan. Safety rules run first.

06 · THREE ENGINES, ONE SWITCH

Choose privacy, speed, or no internet at all

Engine	Runs on	Text goes to	Read a week	Coach answer
Local AI Qwen 2.5 3B	Your laptop	Nowhere	30–90 s	~20 s
Groq gpt-oss-120b	Groq’s cloud	Groq	2–4 s	~1 s
Offline rules Tempo’s parser	Your browser	Nowhere	Instant	Instant

Measured in the real app. If an engine fails, Tempo falls back to offline rules.

The AI never has the last word



- Tempo works out the dates from your words
- Impossible values are rejected
- Tasks the AI drops are recovered by rules
- Every “quote” must appear in what you typed
- “Moved” updates the old task, never a duplicate
- Guesses are yellow; missing details become questions

It learns how long work takes *you*

$$m = \exp\left(\left(\Sigma \ln r + k \cdot \ln m_0 \right) \div (n + k) \right)$$

- r = actual ÷ planned minutes, per session
- m_0 = your overall habit · k = 3
- Few sessions: trust the typical time
- More sessions: trust what really happened

DEMO STUDENT · MATH

45m → 1h

5 past Math sessions ran about 30% long.

Likely 55m–1h 5m · confidence high

A plan must never double-book you, run past your bedtime, or quietly drop work. A chatbot can't promise that. An algorithm can.

So the AI reads and explains. The algorithm schedules, and our tests check every plan.

Experiment E1: [we asked a chatbot to plan the judges' week; it broke X rules]

We tested it, and it broke

WHAT THE MODEL RETURNED

~~Chemistry test: Wed 18 Nov~~
"Next Friday" isn't a Wednesday

~~English essay: Tue 10 Nov~~
Monday is the 9th

~~Task: "Robotics practice"~~
It's an activity, not homework

~~"I only have 2 hours on weekdays"~~
The student never said this

WHAT THE STUDENT SEES AFTER TEMPO'S CHECKS

Chemistry test: **Fri 13 Nov**
Corrected, and flagged to confirm

English essay: **Mon 9 Nov, 2h**
Estimate kept on the right task

Robotics: **Tue & Thu 4–5:30pm**
Once, as an activity

Study time: **unchanged**
Invented sentence ignored

The fix: every quote is checked against what was typed, and tested rules do the date maths. Now an automated test.

Tested on every change, and on the live site

22

automated tests, run before every deploy

7/7

Stage-4 judge scenarios covered by tests

9

problem → change → result stories, written up

2.4s

for Groq to read a week on the live site

Every plan is checked: no day over its limit, no clash with an activity, nothing after a deadline or past bedtime.

Built for students, not for engagement

Private by default

A nickname, not a name. Data stays in your browser.

Won’t do your homework

“Write my essay” gets a refusal and a plan.

Kind when it’s hard

Low-energy days, crunch warnings, a counsellor.

Honest about limits

Ranges, confidence, and a why for every session.

You stay in charge

You confirm every task and pick every trade-off.

Accessible

Keyboard, large text, contrast, voice, read-aloud.

What still needs work

Known limits

- New students start with typical times until about 5 sessions per subject
- The rules miss rare phrasing like “the Thursday after next”
- It plans time, not how well you studied
- Data lives in one browser: private, but no sync

Next

- Test with 5–8 real students: [results]
- Ask “how confident do you feel?” after exam prep
- Compare the AI engines on classmates’ real sentences
- Optional private sync between phone and laptop

Try it *now*.

tempo-study-planner.netlify.app

The judges' week is already loaded. Change something and watch it re-plan.

Code: github.com/hexa3/tempo

[Team names] · [School] · Questions?

